

Celebrate Success. *Share It with Others*

Can there be a better way of remembering the tsunami that hit Japan's nuclear reactor and the stoic bravery of the people who survived this disaster than designing a Geiger counter that's built into a mobile phone, to constantly monitor and feed radiation information to an open sensor network? Can there be a better way of celebrating your own success with teaching robotics than sharing the tools you used with others? Read about these and other interesting open source innovations in this episode of Innovation...

Open source radiation detector

<http://www.bunniestudios.com/blog/?p=2218>



Following the terrible earthquakes that shook Japan in March 2011, radiation concerns increased and the international sales of Geiger counters that measure nuclear radiation soared. As companies like International Medcom ran out of stock, they made it a policy to first provide the instruments they had to those who could share the readings with others. At this time, Safecast—an open sensor network and a terrific example of the open source philosophy—was formed to collect as much radiation data as possible and share it with others. This is when open source advocate Dr Andrew ‘Bunnie’ Huang created a reference design for a compact Geiger counter that people could carry around, and record data with, all the time.

The open twist: The device revolves around a LND 7317 pancake Geiger tube, a large sensor capable of detecting multiple kinds of harmful radiation including Alpha, Beta and Gamma particles. Since Dr Huang

wanted to make sure that people would carry the device all the time, he decided to design the Geiger counter into a mobile phone battery pack—something that is sure to accompany people all the time! So, the design turned out to be a battery pack that can charge a smartphone, but also incorporates a Geiger tube, an LED flashlight (handy in an emergency when there is no power), and some logging circuitry. The Geiger counter would upload its log data to the Safecast network every time the user plugged in the phone to charge. Dr Huang open sourced the designs, thereby enabling companies like International Medcom to manufacture and distribute the devices.

Create your own robot with Multiplo

<http://multiplo.org/>

Multiplo is an open source system to design and build things, especially robots, in an easy way. The founders were teachers, who were upset by the lack of flexible teaching



aids for robotics. Not satisfied by what was available in the market, they started prototyping robots by laser-cutting acrylic and wiring some breadboard to it. Soon, they were